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EM27 Retrieval Pipeline: Automated EM27/SUN Data Processing

Moritz Makowski [1¶], **Frank Hase** [2], **Friedrich Klappenbach** [1], **Andreas Luther** [1], **Lena Feld** [2], **Marlon Müller** [1], **Vyas Giridharan** [1], **Catherine Fait** [3], **and Jia Chen** [1¶]

1 Technical University of Munich (TUM), Munich, Germany **2** Karlsruhe Institute of Technology (KIT), Karlsruhe, Germany **3** Finnish Meteorological Institute, Sodankylä, Finland ¶ Corresponding author

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Software

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Summary

The EM27/SUN (Gisi et al., 2011; Hase et al., 2016) is an FTIR spectrometer used to perform solar absorption measurements to derive column-averaged atmospheric concentrations of greenhouse gases (GHGs). In support of ESA, the COllaborative Carbon Column Observing Network (COCCON) (Alberti et al., 2021; Frey et al., 2019; Herkommer et al., 2024) has been established to offer a framework for the operation and data analysis of the EM27/SUN, ensuring the generation of fiducial GHG observations. The process of estimating the column-averaged concentration of GHGs from the interferograms recorded by the EM27/SUN is called retrieval. The retrieval algorithm established in the COCCON community is PROFFAST (Hase et al., 1999; Hase, 2023; Sha et al., 2020). PROFFASTpylot (Feld et al., 2024) is an interface around PROFFAST that significantly reduces the complexity of running PROFFAST retrievals.

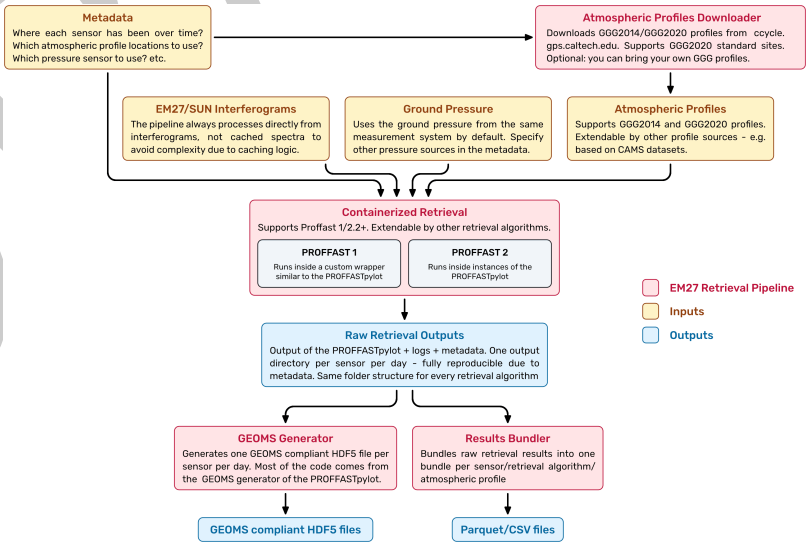

Figure 1: Modules of the EM27 Retrieval Pipeline.

The Munich Urban Carbon Column network (MUCCnet) consists of 5 EM27/SUN spectrometers deployed in and around Munich, Germany (Aigner et al., 2023; Dietrich et al., 2021; Heinle & Chen, 2018). Due to its autonomous operation since 2019, our research group generates a significant amount of EM27/SUN data (see Table 1). Processing this data in an automated and scalable way required us to extend the functionality of the PROFFASTpylot. Under the hood, we use the PROFFASTpylot to ensure consistency with the COCCON community. The

EM27 Retrieval Pipeline is complementary to the PROFFASTpylot, but the more EM27/SUN measurement data one has to retrieve, the more benefits come from this higher degree of automation. While not implementing 100% of the flexibility the PROFFASTpylot provides, this pipeline is flexible enough to be used by any EM27/SUN setup, adhering to the COCCON standards.

We made the following architectural decisions to achieve this higher degree of automation and ease of use:

- **Configuration:** JSON configuration files are used to specify input/output paths, which date range to retrieve, which retrieval algorithm to use, and so on. These files are validated against JSON schemas (Pydantic, 2025; Wright et al., 2022) representing a strict set of rules for their structure and content. A misconfiguration will be detected immediately at program start, as opposed to only failing when the code tries to access invalid parts of the configuration. The JSON schema is rendered as an API reference in the documentation, enabling us to manage and adequately document a large number of configuration parameters.
- **Metadata Management:** Metadata (where/when/how each sensor was set up) is provided as JSON files – again, validated against a documented JSON schema. These files can be stored locally or pulled from a GitHub repository, enabling centralized, version-controlled tracking of sensor deployments. We believe all metadata systems used in different EM27/SUN teams can be easily translated into the schema this pipeline establishes. Our metadata schema design largely agrees with the schema proposed in Zeeman et al. (2024).
- **Parallelization and Containerization:** While the PROFFASTpylot parallelizes the individual steps of PROFFAST, the pipeline parallelizes the whole retrieval process by running one isolated retrieval job per sensor per day. Every result folder maintains the structure of an individual PROFFASTpylot run, but also stores all configuration files required to fully reproduce the output in this folder.
- **Algorithm Support:** While each tagged release of the PROFFASTpylot supports a specific range of PROFFAST versions, the EM27 Retrieval Pipeline keeps support for all implemented retrieval algorithms and versions – the user simply has to specify which algorithm and version to use for a job. Furthermore, we added our own interface for PROFFAST 1 retrievals. The pipeline's architecture is independent of the underlying retrieval algorithm; hence, it could be extended to support GFIT (Connor et al., 2016; Zeng et al., 2021) or other algorithms.
- **Caching/Implicit Scheduling:** Users do not have to define an explicit queue of jobs, but instead define a date range and a list of sensors to include. The downloading/retrieval/bundling will decide which jobs to run. I.e., the pipeline only requests atmospheric profiles that have not been downloaded yet, and it only runs retrievals for data with no existing retrieval result.
- **Testing:** The codebase is statically typed and checked with strict MyPy (Python, 2025) and Pyright (Microsoft, 2025), and includes unit- and end-to-end tests (using PyTest (Krekel et al., 2024)) covering all major functions and retrieval scenarios.

Furthermore, the pipeline adds features required to produce long-term EM27/SUN datasets:

- **A fully automated interface to obtain Ginput data:** The atmospheric profiles downloader automates requests for atmospheric profiles in the GGG2014 or GGG2020 format from a Ginput server (J. Laughner et al., 2021; J. L. Laughner et al., 2023; TCCON, 2025).
- **Bundling of retrieval results:** The raw retrieval outputs are distributed over hundreds or thousands of folders. The bundling script of the pipeline merges these outputs into one file per sensor, retrieval algorithm, and atmospheric profile, which is significantly more suitable for distribution.

Figure 1 shows the current building blocks of the EM27 Retrieval Pipeline. For the full feature set of the pipeline, please refer to its GitHub repository at github.com/tum-esm/em27-

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retrieval-pipeline and documentation. For example, starting with version 1.8, the pipeline extracts important parameters from the given OPUS files using the OPUS-file reader of the `tum-esm-utils` library (Makowski et al., 2025).

Statement of Need

Human-induced climate change is one of the most pressing issues of our time, as it puts the stability of our Earth system at risk. Our civilization is built on that stability. Anthropogenic GHG emissions, as well as natural sources and sinks of GHGs, have to be quantified to create targeted reduction policies, measure policy effectiveness, and identify new emission sources (Calvin et al., 2023).

The EM27/SUN is widely used to achieve these GHG flux estimations (Butz et al., 2017; Chen et al., 2016; Dietrich et al., 2021; Doc et al., 2025; Forstmaier et al., 2023; Hase et al., 2015; Jones et al., 2021; F. Klappenbach et al., 2015; Luther et al., 2019, 2022; Stauber et al., 2025; Tu et al., 2022; Vogel et al., 2019). Unlike the HR125 used by TCCON (Toon et al., 2009), the EM27/SUN can be transported and deployed at different and remote locations without much effort. Since 2016, the instruments of our permanent urban sensor network MUCCnet have been deployed at 3 locations in San Francisco (Friedrich Klappenbach et al., 2021), 4 in Hamburg (Forstmaier et al., 2023), 4 in Vienna (Luther et al., 2023), 12 in Poland (Luther et al., 2019, 2022), and 11 locations in Munich (Chen et al., 2018; Dietrich et al., 2021).

Table 1 shows the data output of our automated setup of MUCCNet (Aigner et al., 2023; Dietrich et al., 2021; Heinle & Chen, 2018) in 2024. Figure 2 shows the XCO2 timeseries of the MUCCnet instruments since September 2019. Running the retrievals for various instrument deployments and keeping track of all deployments over time requires an organizational system that this pipeline provides.

Table 1: Measurement coverage of MUCCnet in 2024.

Sensor	Measurements	Days with at least 3h Coverage	Avg. Daily Coverage
ma (SN061)	368,643	157	4h 38m
mb (SN086)	322,804	141	4h 27m
mc (SN115)	428,582	160	4h 41m
md (SN116)	416,638	151	4h 32m
me (SN117)	431,110	130	3h 53m

To enable a large number of long-term observations with EM27/SUN spectrometers, such as the networks of COCCON-Spain (García et al., 2024) and GEMINI-UK (Humpage et al., 2024), an improvement in automation is crucial. The EM27 Retrieval Pipeline addresses this.

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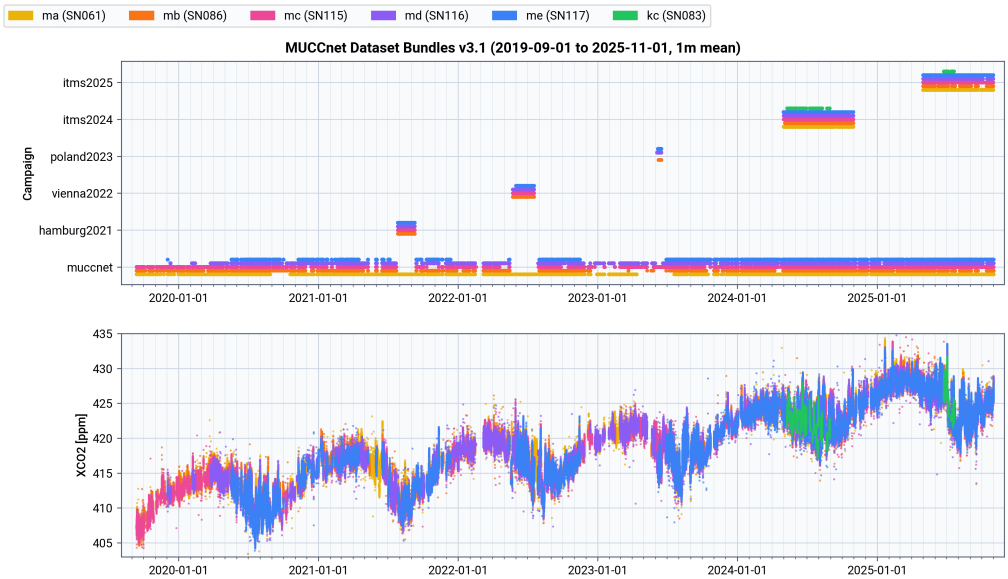

Figure 2: XCO2 of the MUCCnet instruments from 2019-09 to 2025-07

Research Enabled by the EM27 Retrieval Pipeline

The pipeline was first established in 2021 (Rißmann et al., 2022) and has been used for our entire EM27/SUN dataset since 2022 (Chen et al., 2024; Chen, Stauber, et al., 2025; Friedrich Klappenbach et al., 2024; Löw, Feld, et al., 2025; Luther et al., 2023; Stauber et al., 2025; Tang et al., 2024). Multiple journal publications using MUCCnet for inverse modeling are currently in preparation. The data generated by MUCCnet and the EM27 Retrieval Pipeline has been published to the ICOS Carbon Portal (Makowski et al., 2023, 2024) and the EVDC Portal (Chen, Makowski, et al., 2025a, 2025b; Löw, Makowski, et al., 2025) and is a cruicial data input for the GHG modeling approaches developed with the grants listed below.

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